


# Comunicarea in sisteme distribuite

Comunicari sincrone si asincrone, tranzitorii si persistente.

**[Distributed Systems, A.S.Tanenbaum 2003]**




# Comunicarea in sisteme distribuite

- Comunicarea este bazata pe schimb de mesaje
- Nivelul transport este "cheia" ierarhiei
  - ofera o interfata uniforma (socket) pentru programatori
  - furnizeaza conexiuni sigure (ex. TCP)
  - suporta comunicarea multicast
  - limitare - mesajele sunt simple siruri de octeti (nu date structurate)
- Implementat ca TCP, UDP, RTP, T/TCP

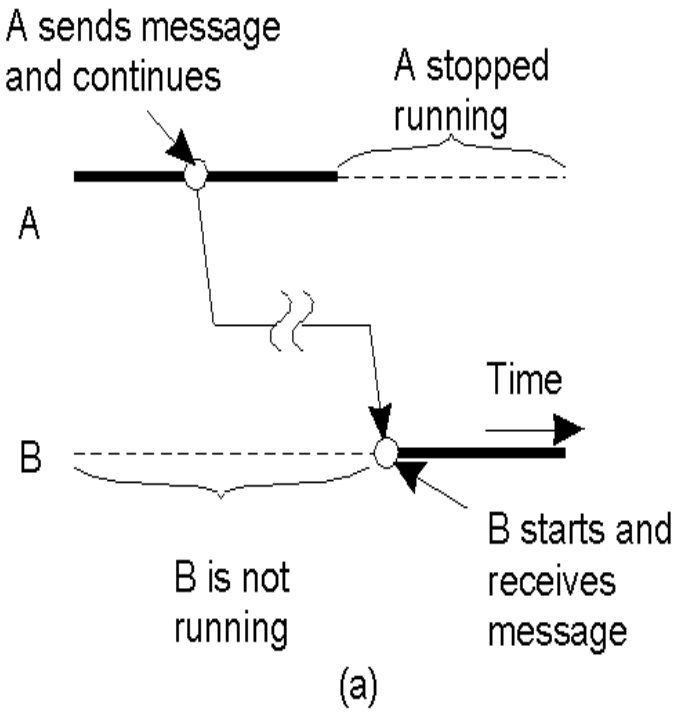



# Modele de comunicare peste nivelul transport

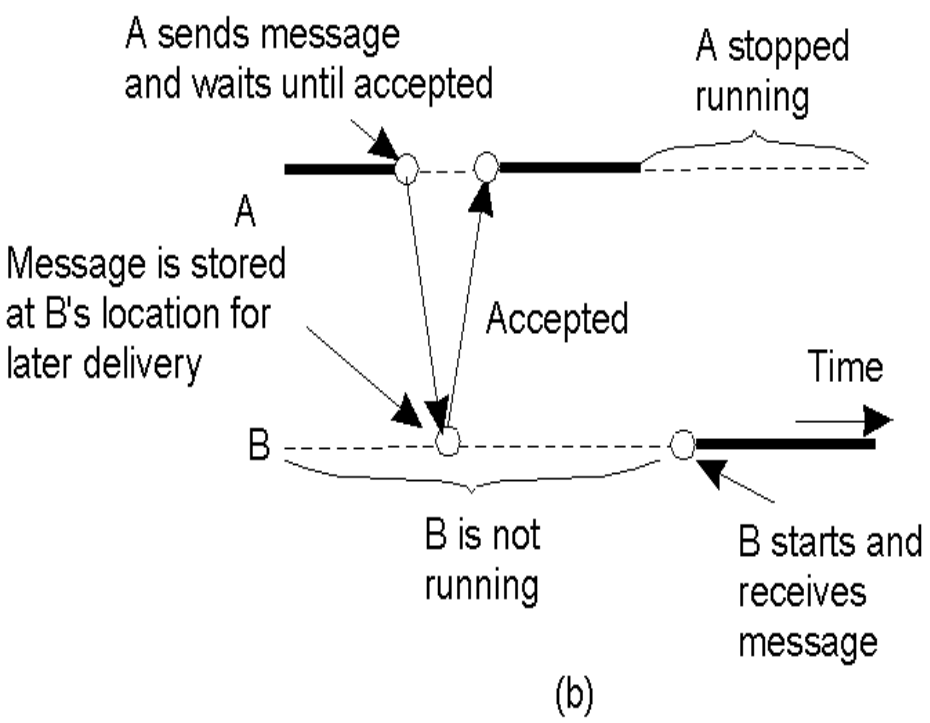
- Protocoale pentru **modelul REST** (HTTP)
- **Extensii ale apelului de procedurl**
  - apeluri de proceduri la distanta (RPC)
  - invocari de metode la distanta (RMI)
  - Simple Object Access Protocol (SOAP)
- **Alte forme ???**
  - comunicarea de grup
  - publish – subscribe (in sisteme bazate pe evenimente)
  - cozi de mesaje
  - multimi de tupluri (tuple spaces)
  - memorie distribuita partajata

# Comunicare persistenta

## asincrona (a)




## sincrona (b)

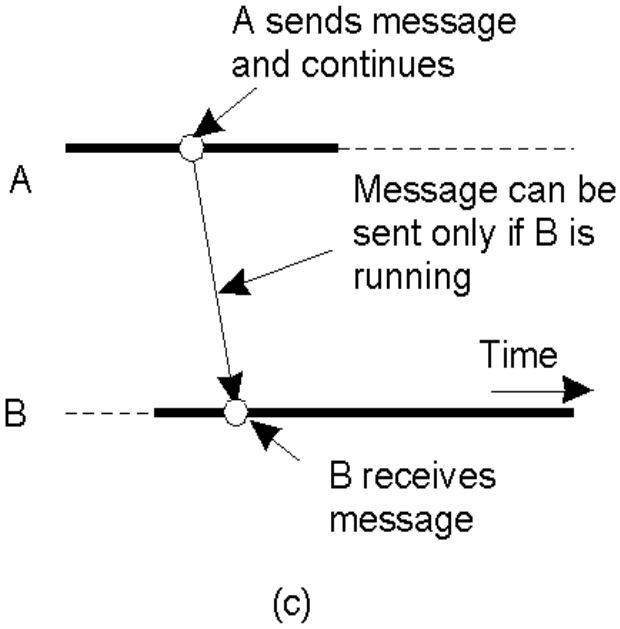



**Un mesaj trimis este pastrat de middleware pana la livrarea lui la receptor**

# Comunicare tranzitorie

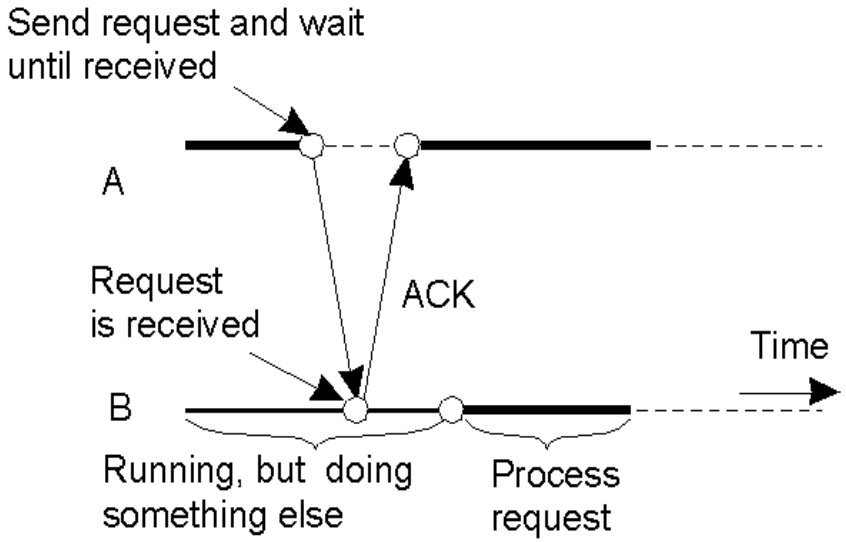
Un mesaj este pastrat de middleware doar cat timp aplicatiile transmitatoare si receptoare sunt in executie

**asincrona**




(c)

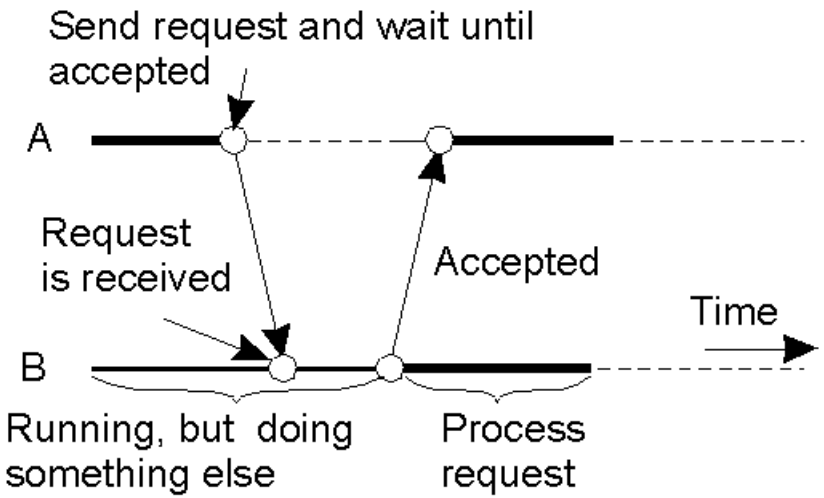
**sincrona la receptie**




(d)

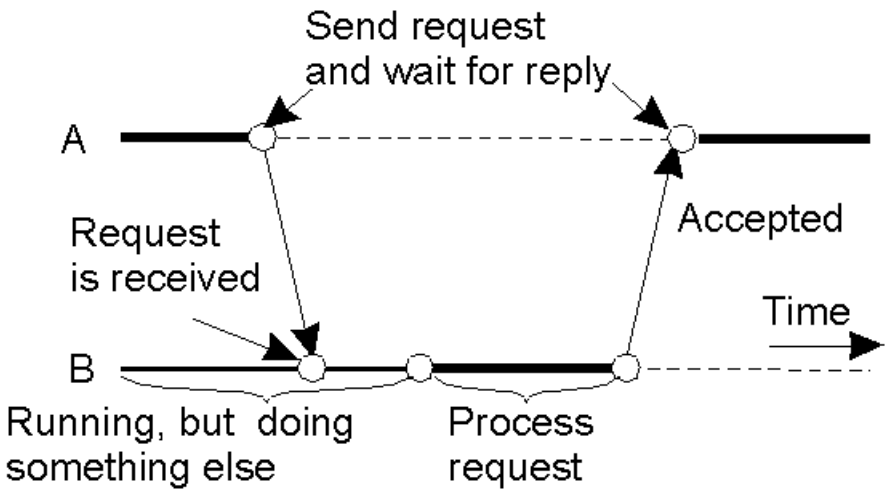
# Comunicare tranzitorie

## sincrona la livrare




(e)

## sincrona la raspuns




(f)